MINISTRY OF EDUCATION, CULTURE AND RESEARCH OF THE REPUBLIC OF MOLDOVA

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REPORT

Laboratory Works No.5 and No.6

The File System with Your Own Hands

Inside the xv6 File System

Operating Systems

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UTM Chișinău – 2026

Purpose of the Laboratory Works

The purpose of Laboratory Work No.5 is to investigate the Linux file-system interface through direct experiments with paths, file types, inodes, links, permissions, storage allocation, mounted devices, and the /proc pseudo-file system. Laboratory Work No.6 continues the same study inside xv6, relating user-visible operations to on-disk structures and kernel code, and implementing two small utilities: mystat and find.

Objectives

1. Distinguish regular files, directories, symbolic links, character devices, and block devices. 2. Explain inode identity, link counts, symbolic-link behavior, directory entries, and sparse-file allocation.
3. Interpret Linux mode bits, umask, directory permissions, set-user-ID, and the sticky bit. 4. Inspect disks, mounts, CPU, memory, process metadata, and file descriptors. 5. Examine xv6 stat, dinode, directory-entry, unlink, and block-reclamation logic. 6. Implement and validate xv6 user programs that display metadata and recursively search directories.

Experimental Environment

The Linux experiments shown here were run on Ubuntu as root. The file-type commands were executed in /root/os-lab5; later tests were repeated in /root/os-lab5-repeat. Because root bypasses ordinary permissions, the permission screenshots do not demonstrate access denial for an ordinary user. xv6-riscv was run in QEMU.

Report Structure

Part I presents the Linux experiments required by Laboratory Work No.5. Part II moves inside xv6 for Laboratory Work No.6. The appendices contain the complete source code of the two implemented utilities, and the report ends with a joint critical analysis and conclusion.

1. Laboratory Work No.5 — The File System with Your Own Hands

1.1 Paths and File Types

Linux exposes several object types through one pathname interface. The first character printed by ls -l identifies the type: d for directory, - for regular file, l for symbolic link, c for character device, and b for block device. The /bin pathname is a symbolic link to /usr/bin on this Ubuntu system; /dev/null is a character device; and /tmp is a directory with the sticky bit set.

```
root@ubuntu-4gb-hel1-1:/# mkdir -p ~/os-lab5
root@ubuntu-4gb-hel1-1:/# cd ~/os-lab5
root@ubuntu-4gb-hel1-1:~/os-lab5# whoami
root
root@ubuntu-4gb-hel1-1:~/os-lab5# pwd
/root/os-lab5
root@ubuntu-4gb-hel1-1:~/os-lab5# printf '\n=== L5-01 FILE TYPES ===\n'

=== L5-01 FILE TYPES ===
root@ubuntu-4gb-hel1-1:~/os-lab5# ls /
app  bin.usr-is-merged  cdrom  etc   lib    lib.usr-is-merged  media  opt      proc  run   sbin.usr-is-merged  srv  tmp  var
bin  boot               dev    home  lib64  lost+found         mnt    os-lab1  root  sbin  snap                sys  usr
root@ubuntu-4gb-hel1-1:~/os-lab5# ls -ld /etc /etc/passwd /bin /dev/null /tmp
lrwxrwxrwx   1 root root     7 Apr 22  2024 /bin -> usr/bin
crw-rw-rw-   1 root root  1, 3 Sep 28 13:52 /dev/null
drwxr-xr-x 119 root root 12288 Oct 24 06:49 /etc
-rw-r--r--   1 root root  1798 Oct 12 12:41 /etc/passwd
drwxrwxrwt  11 root root  4096 Oct 26 11:24 /tmp
root@ubuntu-4gb-hel1-1:~/os-lab5# file /usr/bin/ls /etc/passwd /bin
/usr/bin/ls: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=44900829b39271d878615772bb2a6ffa4
22f0891, for GNU/Linux 3.2.0, stripped
/etc/passwd: ASCII text
/bin:        symbolic link to usr/bin
root@ubuntu-4gb-hel1-1:~/os-lab5# readlink -f /bin
/usr/bin
root@ubuntu-4gb-hel1-1:~/os-lab5# printf '\nCharacter device example:\n'
find /dev -maxdepth 1 -type c -print | head -1

Character device example:
/dev/zfs
root@ubuntu-4gb-hel1-1:~/os-lab5# printf '\nBlock device example:\n'
find /dev -maxdepth 1 -type b -print | head -1

Block device example:
/dev/sr0
root@ubuntu-4gb-hel1-1:~/os-lab5#
```

Figure 1: Linux path types and examples of character and block devices

Observation

A character device transfers a stream of bytes, while a block device is addressed in blocks and normally backs a file system. Symbolic links are distinct objects whose contents are pathnames; therefore /bin and /usr/bin refer to the same directory after pathname resolution.

1.2 Directory Size and Disk Usage

The size shown for a directory entry is metadata storage, not the recursive size of its contents. The du command traverses the hierarchy and totals allocated blocks. In the new measurement, /var/log used about 1.3 GiB and /usr used about 3.5 GiB, making /usr about 2.2 GiB larger.

```
$ ls -ld /var/log /usr
$ du -sh /var/log /usr
```

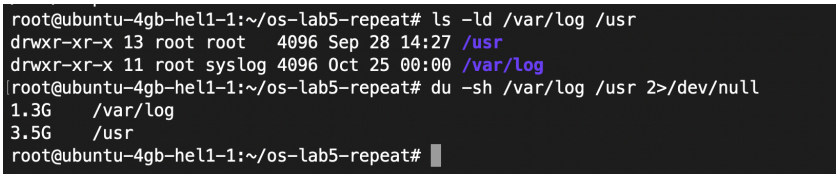


Figure 2: Directory metadata size compared with recursive disk usage

1.3 Inodes and Hard Links

An inode stores metadata and block addresses; a directory maps names to inode numbers. The names notes.txt and hard.txt shared inode 255785 after the hard link was created. Its link count rose from one to two. Removing notes.txt reduced the count, while hard.txt continued to refer to the same data.

```
$ echo hello > notes.txt
$ ls -li notes.txt
$ ln notes.txt hard.txt
$ ls -li notes.txt hard.txt
$ rm notes.txt
$ cat hard.txt
```

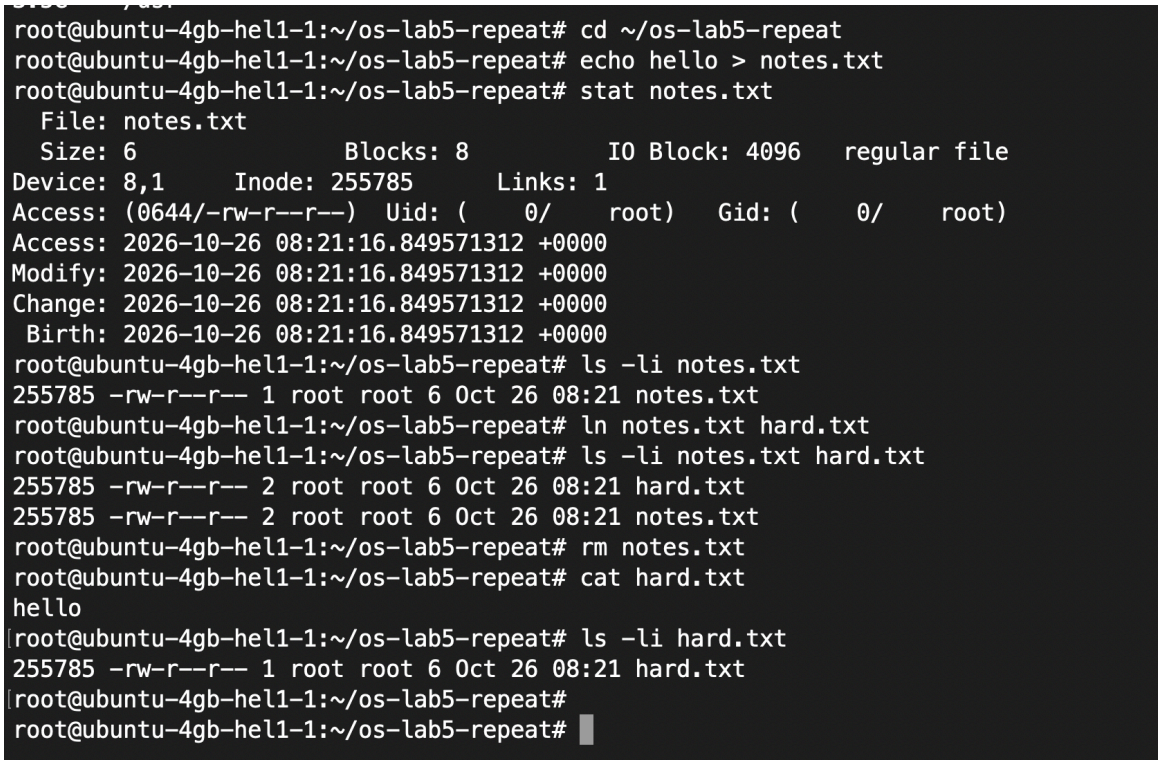


Figure 3: One inode referenced by two hard links

1.4 Symbolic Links and Directory Link Counts

The symbolic link soft.txt had its own inode and stored the eight-character target path hard.txt. Once hard.txt was removed, dereferencing soft.txt failed even though the symbolic-link object remained. A newly created directory had link count two for its parent entry and its own dot entry. Adding subdirectories x and y increased the parent directory's count from two to four because each child contains a .. entry referring to the parent.

```
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ln -s hard.txt soft.txt
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -li hard.txt soft.txt
255785 -rw-r--r-- 1 root root 6 Oct 26 08:21 hard.txt
256217 lrwxrwxrwx 1 root root 8 Oct 26 08:23 soft.txt -> hard.txt
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# cat soft.txt
hello
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# rm hard.txt
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# cat soft.txt
cat: soft.txt: No such file or directory
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -l soft.txt
lrwxrwxrwx 1 root root 8 Oct 26 08:23 soft.txt -> hard.txt
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# mkdir d
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -ld d
drwxr-xr-x 2 root root 4096 Oct 26 08:23 d
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# mkdir d/x d/y
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -ld d
drwxr-xr-x 4 root root 4096 Oct 26 08:23 d
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat#
```

Figure 4: Broken symbolic link and directory link-count growt

1.5 Logical Size versus Allocated Space

File length and allocated space are separate concepts. The two-byte file one.txt occupied one 4 KiB allocation unit. truncate created big.img with a logical length of 1 GiB but no data blocks, so du reported zero allocated space. This sparse representation is useful for disk images and large files containing extensive zero-filled regions.

```
$ echo x > one.txt
$ ls -l one.txt
$ du -h one.txt
$ truncate -s 1G big.img
$ ls -lh big.img
$ du -h big.img
```

```
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# echo x > one.txt
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -l one.txt
-rw-r--r-- 1 root root 2 Oct 26 08:26 one.txt
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# du -h one.txt
4.0K    one.txt
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# truncate -s 1G big.img
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -lh big.img
-rw-r--r-- 1 root root 1.0G Oct 26 08:26 big.img
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# du -h big.img
0       big.img
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat#

root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# stat -c 'logical size: %s bytes; allocated blocks: %b; block unit: %B bytes' big.img
logical size: 1073741824 bytes; allocated blocks: 0; block unit: 512 bytes
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat#
```

Figure 5: Logical length and allocated space of a sparse file

1.6 File Permissions and umask

These checks were run as root. Mode 000 removed ordinary permission bits from s.txt, but root could still read it; this run therefore does not demonstrate denial for an unprivileged account. chmod u+r restored the owner read bit. With umask 0002, new files and directories received modes 664 and 775.

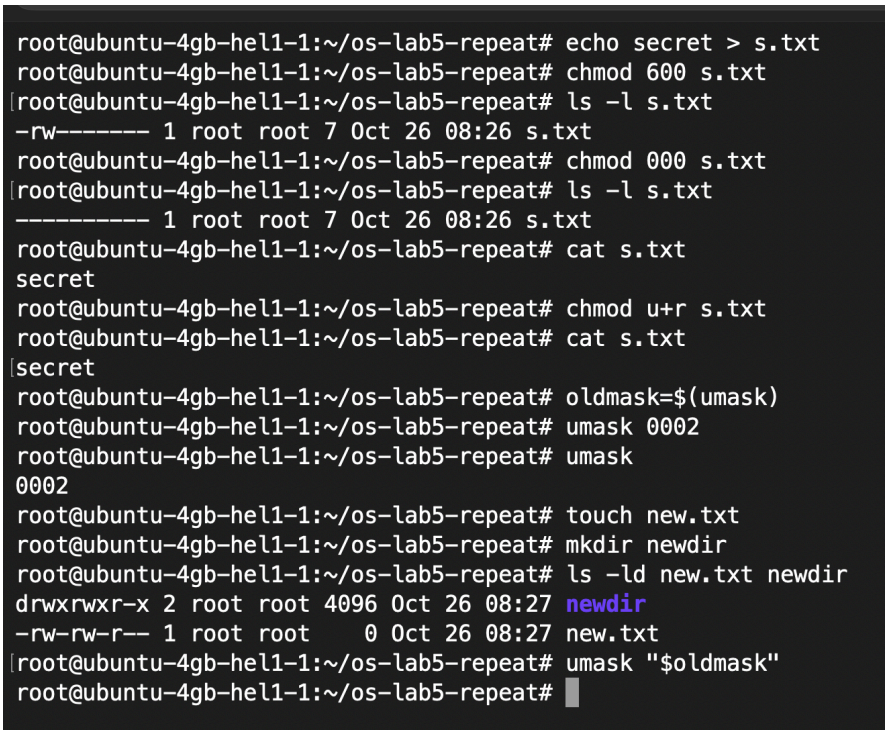


Figure 6: Permission changes as root and umask-derived modes

Recommended Modes

600 is appropriate for private keys and user-only secrets; 644 for ordinary documents readable by others; 700 for private directories or scripts; and 755 for programs and public directories that others must traverse. The least permissive mode that still supports the intended use should be selected.

1.7 Directory Permissions and Special Bits

Directory read permission permits listing names, while execute permission permits traversal. In this run, both ls box and cat box/f succeeded after restrictive mode changes because the commands ran as root. These results do not establish how an ordinary user would be affected.

The set-user-ID bit on /usr/bin/passwd permits operations with the file owner's effective identity. The sticky bit on /tmp limits deletion of other users' files. Root can read /etc/shadow; its contents are excluded from the figure to avoid exposing credential hashes.

```
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# mkdir box
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# echo hi > box/f
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# chmod u=r,go= box
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -ld box
dr-------- 2 root root 4096 Oct 26 08:27 box
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls box
f
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# cat box/f
hi
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# chmod u=x,go= box
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -ld box
d--x------ 2 root root 4096 Oct 26 08:27 box
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls box
f
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# cat box/f
hi
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# chmod 755 box
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -l /usr/bin/passwd
-rwsr-xr-x 1 root root 64152 May 30  2024 /usr/bin/passwd
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -ld /tmp
drwxrwxrwt 11 root root 4096 Oct 26 08:27 /tmp
```

Figure 7: Directory read/execute semantics, set-user-ID, and sticky bit

1.8 Disks and Mounts

The root filesystem was ext4 on /dev/sda1. lsblk showed 1.1 GiB available and 93% use; df -h / showed 38 GiB total, 35 GiB used, 1.1 GiB available, and 98% use. The tools account for reserved blocks differently. The FAT32 EFI partition on /dev/sda15 was mounted at /boot/efi.

```
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# lsblk -f
NAME    FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1  ext4   1.0         64f6fb02-4fed-4836-a33b-86e8993afdfa    1.1G    93% /
├─sda14
└─sda15 vfat   FAT32       227C-A484                             251.9M     0% /boot/efi
sr0
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        38G   35G  1.1G  98% /
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# findmnt /
TARGET
  SOURCE    FSTYPE OPTIONS
/ /dev/sda1 ext4   rw,relatime
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# mount | grep ' on / '
/dev/sda1 on / type ext4 (rw,relatime)
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat#
```

Figure 8: Block devices, root-file-system usage, and mount information

1.9 The /proc Pseudo-File System

The system exposed two processors and 3,911,572 kB of physical memory, with approximately 2.0 GiB available and no swap configured. /proc/self refers dynamically to the process reading it; consequently head observed its own status and its own file-descriptor table. Descriptors 0, 1, and 2 pointed to the terminal, while descriptor 3 referred to the directory being listed.

```
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# grep -c '^processor' /proc/cpuinfo
2
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# head -3 /proc/meminfo
MemTotal:        3911572 kB
MemFree:          729380 kB
MemAvailable:    2038588 kB
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       1.8Gi       723Mi       5.3Mi       1.5Gi       2.0Gi
Swap:             0B          0B          0B
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# head -5 /proc/self/status
Name:   head
Umask:  0022
State:  R (running)
Tgid:   2152338
Ngid:   0
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat# ls -l /proc/self/fd
total 0
lrwx------ 1 root root 64 Oct 26 08:28 0 -> /dev/pts/0
lrwx------ 1 root root 64 Oct 26 08:28 1 -> /dev/pts/0
lrwx------ 1 root root 64 Oct 26 08:28 2 -> /dev/pts/0
lr-x------ 1 root root 64 Oct 26 08:28 3 -> /proc/2152390/fd
root@ubuntu-4gb-hel1-1:~/os-lab5-repeat#
```

Figure 9: CPU, memory, process status, and open file descriptors in /proc

2. Laboratory Work No.6 — Inside the xv6 File System 2.1 Metadata Structures

xv6 defines three public stat types: T_DIR = 1, T_FILE = 2, and T_DEVICE = 3. struct stat contains the device identifier, inode number, type, link count, and byte size. Its on-disk counterpart, struct dinode, additionally stores major and minor device numbers and an array of direct and indirect block addresses.

```
loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$ printf '\n=== L6-01 STAT TYPES ===\n'; nl -ba kernel/stat.h

=== L6-01 STAT TYPES ===
     1  #define T_DIR    1 // Directory
     2  #define T_FILE   2 // File
     3  #define T_DEVICE 3 // Device
     4
     5  struct stat {
     6    int dev;     // File system's disk device
     7    uint ino;    // Inode number
     8    short type;  // Type of file
     9    short nlink; // Number of links to file
    10    uint64 size; // Size of file in bytes
    11  };
loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$ printf '\n=== L6-01 DINODE FIELDS ===\n'; grep -n -A18 'struct dinode' kernel/fs.h

=== L6-01 DINODE FIELDS ===
32:struct dinode {
33-  short type;               // File type
34-  short major;              // Major device number (T_DEVICE only)
35-  short minor;              // Minor device number (T_DEVICE only)
36-  short nlink;              // Number of links to inode in file system
37-  uint size;                // Size of file (bytes)
38-  uint addrs[NDIRECT + 1]; // Data block addresses
39-};
40-
41-// Inodes per block.
42:#define IPB (BSIZE / sizeof(struct dinode))
43-
44-// Block containing inode i
45-#define IBLOCK(i, sb) ((i) / IPB + sb.inodestart)
46-
47-// Bitmap bits per block
48-#define BPB (BSIZE * 8)
49-
50-// Block of free map containing bit for block b
51-#define BBLOCK(b, sb) ((b) / BPB + sb.bmapstart)
52-
53-// Directory is a file containing a sequence of dirent structures.
54-#define DIRSIZ 14
55-
56-// The name field may have DIRSIZ characters and not end in a NUL
57-// character.
58-struct dirent {
59-  ushort inum;
60-  char name[DIRSIZ] __attribute__((nonstring));
loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$
```

Figure 10: xv6 stat types, struct stat, and struct dinode fields

2.2 Root Directory and Directory Entries

The kernel booted successfully with two harts. xv6 assigned inode 1 to the root directory. Both . and .. in the root refer to inode 1 because the root is its own parent. A new empty directory occupied 32 bytes and contained two 16-byte entries: a two-byte inode number and a fourteen-byte name field.

```
$ ls
$ mkdir emptydir
$ ls emptydir
```

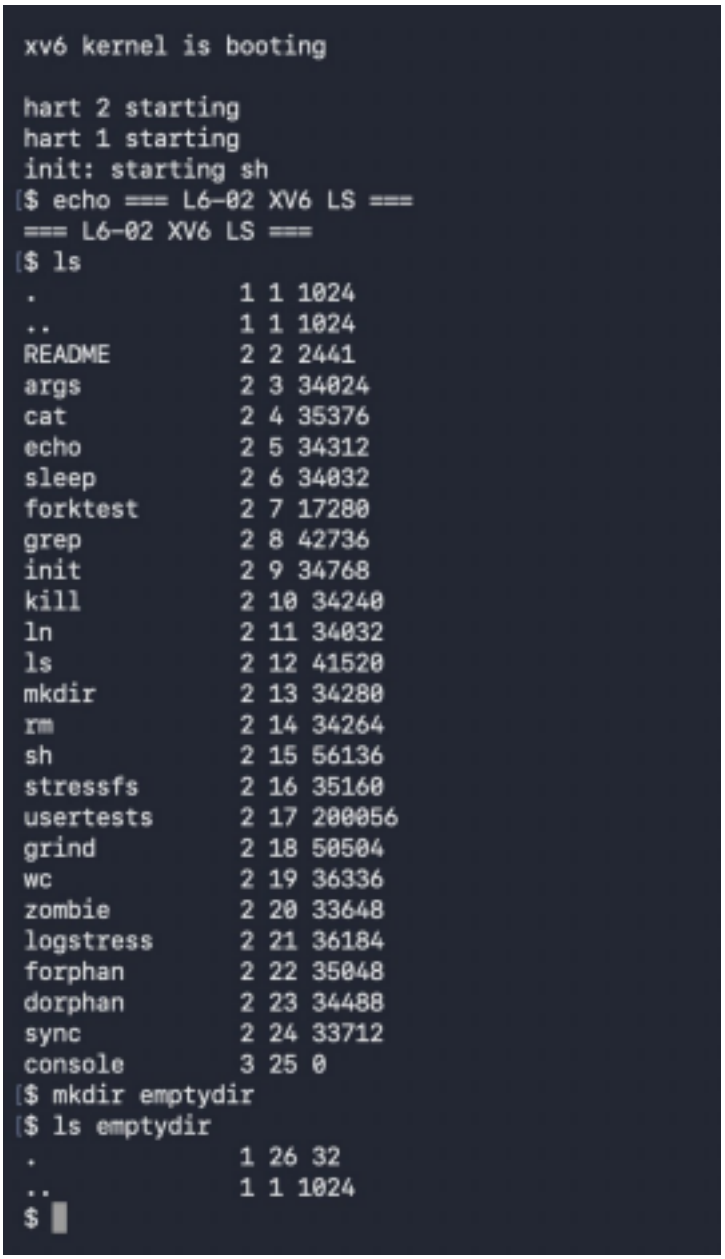


Figure 11: xv6 boot, root listing, and the entries of an empty directory

2.3 Registering User Programs

The programs were added to the UPROGS list as $U/_mystat and $U/_find. This causes their user-space binaries to be linked and copied into fs.img by mkfs, making them available in the xv6 shell after the next build.

```
loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$ python3 -c 'from pathlib import Path; p=Path("Makefile"); s=p.read_text(); marker="\t$U/_cat\\\n"; assert marker in s and "$U/_mystat" not in s and "$U/_find" not in s; p.write_text(s.replace(marker, marker+"\t$U/_mystat\\\n\t$U/_find\\\n", 1))'
loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$ grep -n -B2 -A4 '_mystat\|_find' Makefile
133-    $U/_args\
134-    $U/_cat\
135:    $U/_mystat\
136:    $U/_find\
137-    $U/_echo\
138-    $U/_sleep\
139-    $U/_forktest\
140-    $U/_grep\
loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$ sed -n '132,142l' Makefile
UPROGS=\\$
\t$U/_args\\$
\t$U/_cat\\$
\t$U/_mystat\\$
\t$U/_find\\$
\t$U/_echo\\$
\t$U/_sleep\\$
\t$U/_forktest\\$
\t$U/_grep\\$
\t$U/_init\\$
\t$U/_kill\\$
loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$
```

Figure 12: Makefile UPROGS entries for mystat and find

2.4 mystat and Link Semantics

mystat calls stat for every pathname and translates the numeric type to a readable label. README was a regular file at inode 2 with one link and a size of 2,441 bytes; . was directory inode 1; and console was device inode 27. After ln README r2, both names reported inode 2 and link count two. Removing README left r2 valid with link count one and unchanged contents.

```
xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ echo === L6-04 MYSTAT ===
=== L6-04 MYSTAT ===
$ mystat README . console
README: file, inode 2, links 1, size 2441 bytes
.: directory, inode 1, links 1, size 1024 bytes
console: device, inode 27, links 1, size 0 bytes
$ ln README r2
$ mystat README r2
README: file, inode 2, links 2, size 2441 bytes
r2: file, inode 2, links 2, size 2441 bytes
$ rm README
$ mystat r2
r2: file, inode 2, links 1, size 2441 bytes
$ cat r2 | wc
48 336 2441
$
```

Figure 13: mystat output before and after linking and unlinking README

2.5 Unlink and Block Reclamation

sys_unlink removes the directory entry and decrements ip->nlink, persisting the new value with iupdate. For a directory, the parent's link count is also decremented. The data blocks are not necessarily freed at that instant: iput checks for the final in-memory reference to a valid inode whose nlink is zero, locks it, and calls itrunc to release its data blocks. This separation ensures that an open file remains usable after its final pathname has been removed.

```
[loredana@ubuntu-4gb-hel1-1:~/xv6-riscv$ printf '\n=== L6-05 UNLINK AND BLOCK FREEING ===\n'; grep -n -C10 'nlink--' kernel/sysfile.c

=== L6-05 UNLINK AND BLOCK FREEING ===
168-  }
169-  iunlockput(dp);
170-  iput(ip);
171-
172-  end_op();
173-
174-  return 0;
175-
176-bad:
177-  ilock(ip);
178:  ip->nlink--;
179-  iupdate(ip);
180-  iunlockput(ip);
181-  end_op();
182-  return -1;
183-}
184-
185-// Is the directory dp empty except for "." and ".." ?
186-static int
187-isdirempty(struct inode *dp)
188-{
--
229-    panic("unlink: nlink < 1");
230-  if (ip->type == T_DIR && !isdirempty(ip)) {
231-    iunlockput(ip);
232-    goto bad;
233-  }
234-
235-  memset(&de, 0, sizeof(de));
236-  if (writei(dp, 0, (uint64)&de, off, sizeof(de)) != sizeof(de))
237-    panic("unlink: writei");
238-  if (ip->type == T_DIR) {
239:    dp->nlink--;
240-    iupdate(dp);
241-  }
242-  iunlockput(dp);
243-
244:  ip->nlink--;
245-  iupdate(ip);
246-  iunlockput(ip);
247-
248-  end_op();
249-
250-  return 0;
251-
252-bad:
253-  iunlockput(dp);
254-  end_op();
```

Figure 14: Link-count updates in the xv6 unlink system call

Conclusion and Critical Analysis

The two laboratory works connect the same abstraction at two levels. Linux presents files, directories, devices, and process information through a uniform pathname interface, while xv6 makes the underlying inode, directory-entry, block, and reference-count mechanisms small enough to trace directly. The experiments confirmed that a pathname is not the file itself: hard links share one inode, symbolic links contain another pathname, and unlink removes a name before it necessarily releases storage.

The permission experiments also showed why tests must be performed as an ordinary user. Owner identity chooses the owner mode bits but does not override them, directory read and execute permissions protect different operations, and special bits deliberately change deletion or execution semantics. The sparse-file experiment further demonstrated that logical length is not a reliable measure of physical consumption.

The most useful implementation lesson came from testing find with a directory as the target. The first version compared the basename only in the non-directory branch, so find a b traversed directory b but did not print it. Moving the comparison before the type test corrected the behavior without preventing recursion. This illustrates why a successful build is insufficient: tests must cover regular files, nested paths, hard-linked names, directories, and boundary conditions such as DIRSIZ.

Overall, the Linux observations and xv6 source inspection support one coherent model: directory entries provide names, inodes provide identity and metadata, link counts record persistent namespace references, open references preserve temporary access, and blocks are reclaimed only when both persistent and active references permit it. Both laboratory works were completed successfully, and the final xv6 image verified the required utilities.